

# Day 30 - Euler-Lagrange Equation

**MICHIGAN STATE**  
UNIVERSITY

Welcome to  
PHY 321:  
Classical  
Mechanics

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The future that liberals want



# Reminders

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We proposed a solution to the line problem that involved an error term  $\eta(x)$ , which is a small perturbation to the true path  $y(x)$ . This leads to a perturbed function:

$$Y(x) = y(x) + \alpha\eta(x)$$

where  $\alpha$  is a small parameter.

We proposed that there's a functional  $f(Y, Y', x)$  that depends on a function  $Y(x)$ , its derivative  $Y'(x)$ , and the independent variable  $x$  such that:

$$\int_{s_1}^{s_2} f(Y, Y', x) dx > \int_{s_1}^{s_2} f(y, y', x) dx$$

# Reminders

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By taking the derivative of the functional with respect to  $\alpha$ , we can find the condition for which the functional is stationary (i.e., a minimum or maximum).

$$\left. \frac{d}{d\alpha} \int_{s_1}^{s_2} f(Y, Y', x) dx \right|_{\alpha=0} = 0$$

This (with a lot of math) led us to the following expression:

$$\int_{s_1}^{s_2} \eta(x) \left[ \frac{\partial f}{\partial y} - \frac{d}{dx} \left( \frac{\partial f}{\partial y'} \right) \right] dx = 0$$

## Clicker Question 30-1

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We completed this derivation with the following mathematical statement:

$$\int_{s_1}^{s_2} \eta(x) \left[ \frac{\partial f}{\partial y} - \frac{d}{dx} \left( \frac{\partial f}{\partial y'} \right) \right] dx = 0$$

where  $\eta(x)$  is an arbitrary function. What does this imply about the term in square brackets?

1. The term in square brackets must be a pure function of  $x$ .
2. The term in square brackets must be a pure function of  $y$ .
3. The term in square brackets must be a pure function of  $y'$ .
4. The term in square brackets must be zero.
5. The term in square brackets must be a non-zero constant.

## Clicker Question 30-2

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Returning to the line problem,

$$l = \int_{x_1}^{x_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

here,  $f(y, y', x) = \sqrt{1 + (y')^2}$ , where  $y' = \frac{dy}{dx}$ .

Apply the Euler-Lagrange equation to find the expression for the function  $f(y, y', x)$  in this case. Write your result to find the expression for the term in square brackets:

$$\frac{d}{dx} [?] = 0$$

**Click when you have an answer!**

## Clicker Question 30-3

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With,

$$y' = \pm \sqrt{\frac{c^2}{1 + c^2}}$$

where  $c$  is a constant, the solution expresses a straight line.

1. True and I can prove it!
2. True, but I'm not sure how to prove it.
3. False, I think this is incorrect.
4. I don't know.

## Clicker Question 30-4

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We derived the time that it takes to run from a point on the shore to a point in the water,  $T$ :

$$T = \frac{1}{v_1} (x_1^2 + (y - y_1)^2)^{1/2} + \frac{1}{v_2} (x_2^2 + (y_2 - y)^2)^{1/2}$$

To find the minimal time, what derivative should we take?

1.  $\frac{dT}{dx}$
2.  $\frac{dT}{dy}$
3.  $\frac{dT}{dt}$
4. Something else

## Clicker Question 30-5

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For the brachistochrone problem, the ball moves purely under the influence of gravity. Consider that the ball has moved a vertical distance  $\Delta y$  from rest.

What is the speed of the ball at this point?

1.  $v = gt$

2.  $v = 2g\Delta y$

3.  $v = \sqrt{2g\Delta y}$

4. I'm not sure, but  $< \sqrt{2g\Delta y}$

5. Something else?